TESTING PACK

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How to test

1. Login & Access Control Testing

1.1 Valid Admin Login

1. Enter a valid admin username and password.
2. Expected: User successfully logs into the AdminDashboard without errors.

1.2 Invalid Login

1. Use incorrect password or unknown username.
2. Expected: System shows “Invalid credentials” message.
3. No crash should occur.

1.3 Role Validation

1. Attempt logging in as a student with student credentials.
2. Expected: StudentDashboard should load instead of AdminDashboard.

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2. Sidebar Navigation Testing

2.1 Button Highlighting

1. Click **Home**, **Manage Users**, **Manage Courses**, **Maintenance**, **More**, etc.
2. Expected:
    - The active tab turns cyan (#00BFBF).
    - Other tabs revert to white background.
    - Correct panel is displayed in content area.

2.2 Persistence of Selected Panel

1. Navigate between multiple panels.
2. Expected:
    - No graphical glitch.
    - No overlapping components.

- No exceptions in console.

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3. Maintenance Mode Testing

3.1 Toggle ON Confirmation

1. Click the toggle switch.
2. Expected:
    - A confirmation dialog appears: “Continue” or “Cancel”.
    - If “Cancel” → toggle returns to OFF state.

3.2 Toggle ON Behavior

1. Choose “Continue.”
2. Expected:
    - Toggle turns green.
    - Maintenance banner appears in AdminDashboard.
    - An entry is written to database table **settings** (maintenance_mode = 'ON').

3.3 Toggle OFF Behavior

1. Switch the toggle off.
2. Expected:
    - Banner disappears.
    - Database updates to maintenance_mode = 'OFF'.

3.4 Student Dashboard Reaction

1. Log in as a student while maintenance = ON.
2. Expected:
    - Yellow banner displayed in StudentDashboard.
    - Register/Drop buttons disabled OR produce “Cannot register during Maintenance Mode” message.
    - Student can still view all data.

3.5 Instructor Dashboard Reaction

1. Log in as a instructor while maintenance = ON.
2. Expected:
    - Yellow banner displayed in InstructorDashboard.
    - Grade buttons disabled OR produce “Cannot grade during Maintenance Mode” message.
    - Instructor can still view all data.

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4. User Management Testing

4.1 Load Users

1. Navigate to “Manage Users.”
2. Expected:
    - Table populates correctly.
    - No missing rows.

4.2 Add User

1. Click **Add User**.
2. Fill all fields and save.
3. Expected:
    - Pretty, centered dialog box appears.
    - New user shows in table.
    - Password is hashed properly.
    - Database users and users_auth updated.

4.3 Validation Errors

Test missing fields:

- Empty username
- Empty role
- Duplicate username
  Expected: Clear error messages, no crash.

4.4 Edit User

1. Select a user → click Edit.
2. Change details and save.
3. Expected: Changes appear in table and DB.

4.5 Delete User

1. Select row → delete.
2. Expected: Confirm dialog → row removed → DB updated.
3. Cannot delete users referenced in foreign keys (expect error message).

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5. Course Management Testing

5.1 Load Courses

1. Navigate to “Manage Courses.”

2. Expected:
    - All columns load, including new drop_deadline field.

5.2 Add Course

1. Click **Add Course**.
2. Fill all fields including date fields.
3. Save.
4. Expected: New course appears in table.

5.3 Edit Course

1. Select course → edit.
2. Modify instructor, deadline, etc.
3. Expected: Table and DB update correctly.

5.4 Delete Course

1. Delete a course.
2. If linked to timetable/materials/registrations → expect error message.
3. If not linked → deletion succeeds.

6. Panels & UI Testing

6.1 Responsiveness

1. Resize window.
2. Expected: Panels adjust gracefully without distortion.

6.2 Banner Visibility

1. Toggle maintenance ON/OFF.
2. Navigate between pages.
3. Expected: Banner always stays visible when ON.

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7. Database Consistency Tests

7.1 Admin Setting Reflection

1. Set maintenance ON in DB manually.
2. Restart StudentDashboard.
3. Expected: Banner shows immediately.

7.2 Referential Integrity

Attempt deleting:

- Course referenced in timetable.

- User referenced in registrations.
  Expected: Error shown, app doesn’t crash.

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8. Error Handling Tests

- Disconnect database and try navigating panels.
  Expected: “Database unreachable” type message, not crash.
- Attempt SQL injection in text fields.
  Expected: No SQL error, input sanitized.